

Advances in the pathophysiology and treatment of diabetic peripheral neuropathy

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ABSTRACT

Diabetic peripheral neuropathy (DPN) is a common, debilitating disorder that results in sensory loss, gait dysfunction, propensity to skin ulceration that can lead to limb amputations, and neuropathic pain. DPN has a negative impact on patients' mood and quality of life and is also associated with increased mortality. Disappointingly, aggressive treatment of hyperglycemia does not adequately prevent DPN, particularly in patients with type 2 diabetes, and new, more effective treatments for DPN are urgently needed. Recent decades have witnessed a growing understanding of the risk factors and molecular mechanisms of DPN, such as the importance of axoglial interactions and metabolic derangements beyond hyperglycemia. Despite expanding knowledge, the translation of preclinical insights to treatments that benefit patients remains challenging. This review will cover DPN epidemiology and risk factors, including social determinants of health, DPN pathophysiology and genetics, currently recommended diagnostic and treatment approaches, and available clinical outcomes assessments and biomarkers. Discussion of each of these topics will emphasize recent advances in the field and focus on the challenges and necessary new directions in DPN research.

Introduction

The impact of diabetes and its complications, the rates of which are rapidly increasing, cannot be overstated. Although diabetes associated neurological complications are diverse and can involve both the peripheral and central nervous systems, this review will focus on diabetic peripheral neuropathy (DPN), which affects up to 50% of people living with diabetes.¹⁻⁴ DPN results in sensory loss and gait dysfunction, as well as propensity to foot ulceration that can lead to lower limb amputations. Many patients also experience chronic neuropathic pain that can be difficult to control with drugs. The symptoms of DPN negatively affect daily function and quality of life, resulting in high rates of anxiety and depression.⁵⁻⁸

Alarming, a systematic analysis from the Global Burden of Disease Study in 2021 identified DPN as one of the top five leading neurological causes of cumulative fatal and non-fatal disease burden. Among the varied neurological disorders, DPN also showed the highest increase in disability adjusted life years between 1990 and 2021 (change of 92%).⁹ Furthermore, several large studies (n=7137 and n=155 934) have shown a marked increase in the risk of cardiovascular and all cause mortality (hazard ratio 1.96-2.04 and 1.72, respectively).^{10 11}

The past several decades have seen notable progress in defining risk factors and molecular

mechanisms underlying DPN. Specifically, a growing appreciation for the contribution of metabolic derangements beyond hyperglycemia has paved the way for new treatment directions. Despite these advances, however, no disease modifying treatment is available for DPN beyond optimizing control of glycemic and metabolic risk factors, and the translation of preclinical insights to clinical trials remains a challenge.

This review will cover the epidemiology of and risk factors for DPN, including social determinants of health, the pathophysiology and genetics of DPN, currently recommended diagnostic and treatment approaches, and clinical outcomes assessments and biomarkers. This detailed update aims to capture the complexity of emerging DPN risk factors and biology and highlight current barriers and future research directions in DPN (fig 1).

Sources and selection criteria

We did individual searches for studies on PubMed published between 1 January 2001 and 10 March 2025 by using terms relevant to specific topics pertaining to diabetic peripheral neuropathy. Search terms included biomarkers, clinical outcome measures, clinical trials, diabetes, diabetic neuropathy, diabetic peripheral neuropathy, diagnosis, dyslipidemia, hyperglycemia, low and middle income countries, longitudinal, metabolic

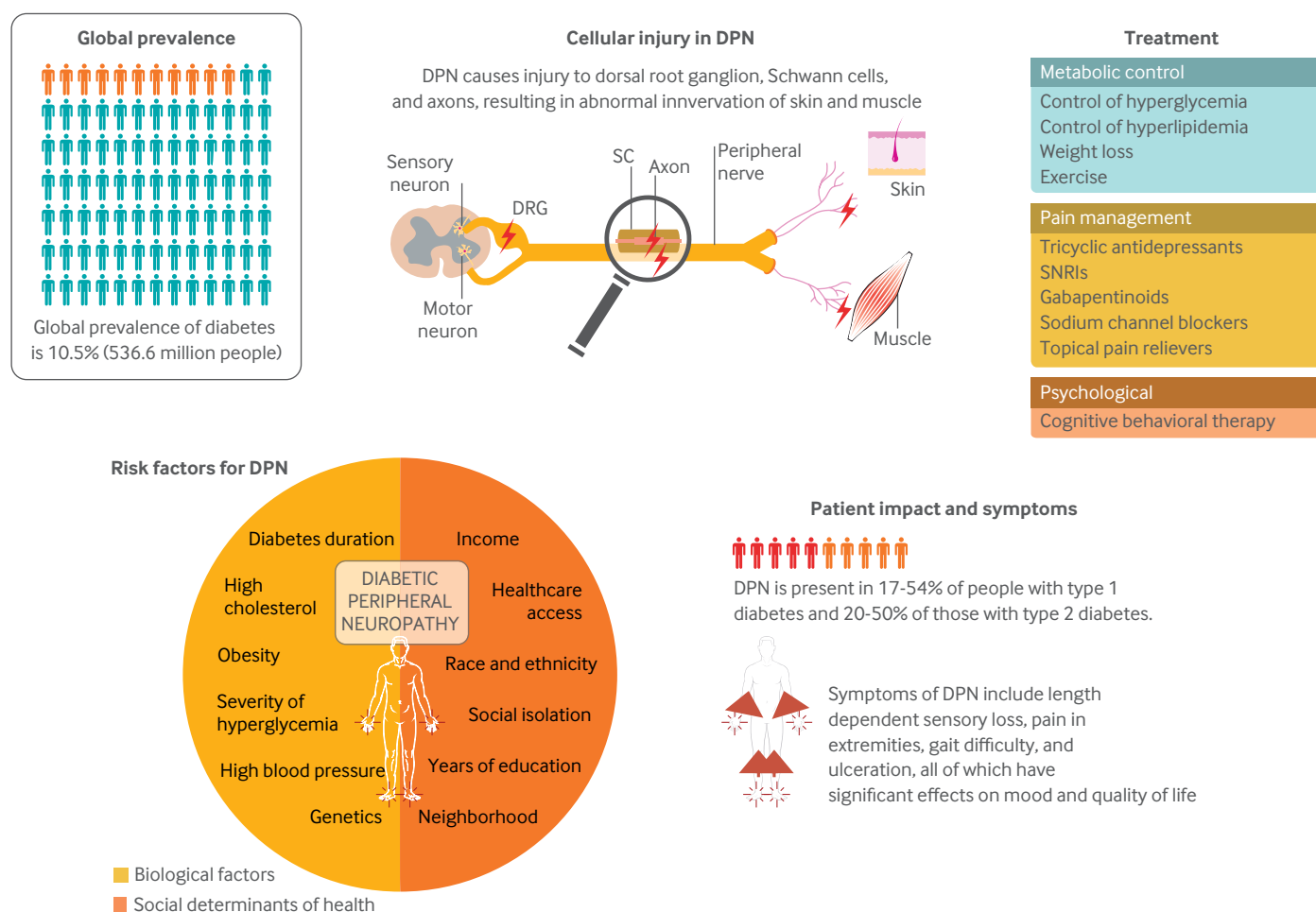


Fig 1 | Overview of diabetic peripheral neuropathy (DPN). The prevalence of diabetes and DPN is high and rapidly increasing. Risk factors for DPN are diverse and heterogeneous, including duration and severity of diabetes, other metabolic risk factors, genetics, and social determinants of health. Cellular injury in DPN is complex, affecting the dorsal root ganglion, axon, and Schwann cell, with subsequent disruption of skin and muscle innervation by peripheral nerves. The clinical presentation of DPN most commonly involves length dependent sensory loss, with neuropathic pain being common but not universal. Treatment approaches remain limited to dietary, surgical, and drug interventions aimed at optimizing glycemic control and other metabolic risk factors, exercise interventions, and oral and topical treatments for neuropathic pain. SNRI=serotonin-norepinephrine reuptake inhibitor

syndrome, mouse models, MRI, pathogenesis, prevalence, quality of life, Schwann cells, social determinants of health, socioeconomic status, treatment, type 1 diabetes, type 2 diabetes, ultrasound, and under-resourced setting. We considered all studies published between 1 January 2001 and 10 March 2025 and hand searched references of key articles for additional articles. We prioritized studies according to the highest level of evidence and time of publication, with focus on systematic reviews, meta-analyses, large high quality observational studies, and consensus statements. We also included published reviews when clinically relevant and key publications outside of our review dates to add historical relevance. We did not include studies that did not have an English or Spanish abstract, case reports, or case series.

Epidemiology

Diabetes is an epidemic worldwide. A study encompassing 219 data sources from broad geo-

graphic regions between 2005 and 2020 across 215 countries estimated a global prevalence at 10.5% (536.6 million people), with an anticipated rise to 12.2% (783.2 million) by 2045.¹² As will be discussed in the “Global issues” section, the reported prevalence of DPN has varied across geographic regions, with a range of 20-50% in people with type 2 diabetes and 17-54% in those with type 1 diabetes.^{2 13 14} In addition to reflecting regional differences in patient demographics, DPN risk factors, and varied clinical practices, this variability reflects the differing diagnostic approaches and definitions of DPN across studies.¹⁵

Alarming, a high prevalence of DPN has been reported in large cohorts of participants with youth onset type diabetes.¹⁶⁻¹⁸ The SEARCH for Diabetes in Youth (SEARCH) cohort study found a prevalence of 22% and 7% among young people with type 2 and type 1 diabetes, respectively (n=1734 for type 1 diabetes; n=258 for type 2 diabetes).¹⁶ A 25.4% increase in the prevalence of neuropathy over 14

years was also observed in the Treatment Options for Type 2 Diabetes in Adolescents and Youth (TODAY) study (n=674).¹⁷

Neuropathic pain is not universal in people with DPN, and the absence of pain may be resulting in underestimations of DPN prevalence. The reported prevalence of DPN associated neuropathic pain has ranged from 5% to 45% in people with type 1 diabetes and from 15% to 57% in those with type 2 diabetes.^{2 13 19-22} A recent meta-analysis of 41 studies involving 36 983 participants with DPN reported a painful DPN prevalence of 41%, and a recent review of global and regional DPN prevalence estimated that 20-40% of patients with type 2 diabetes have painful neuropathy.^{5 23}

Metabolic risk factors

Diabetes: severity and duration

Diabetes is the leading cause of peripheral neuropathy, and both duration and severity of diabetes contribute to the risk of having DPN among people with diabetes.²⁴ Severity of hyperglycemia, often measured by glycated hemoglobin (hemoglobin A_{1c} (HbA_{1c})), has consistently been associated with increased risk of DPN in prospective cohort studies.^{24 25} Furthermore, longer durations of diabetes increase the likelihood of DPN independently of age and HbA_{1c} level.^{24 25}

Outside of HbA_{1c} and duration of diabetes, several studies have assessed how other markers of diabetes are associated with DPN. For example, hyperinsulinemia, measured as high β cell function (homeostatic model assessment (HOMA2-B) and low insulin sensitivity (HOMA2-S) was found to increase the risk of DPN among 3397 Danish adults with type 2 diabetes.²⁶ Studies in China and Germany have found that people with severe insulin deficient diabetes or severe insulin resistant diabetes have increased risk of DPN, compared with other, less severe diabetes subtypes such as mild age related diabetes.^{27 28}

Metabolic syndrome

The metabolic syndrome is a cluster of risk factors including hyperglycemia, abdominal obesity, hypercholesterolemia, hypertriglyceridemia, and hypertension. It has been found to increase the likelihood of DPN in both type 1 diabetes and type 2 diabetes.²⁹

The largest study to date, the Metascreen multicenter diabetes survey, found that having the metabolic syndrome increased the likelihood of having DPN in 7859 people with type 2 diabetes and 638 people with type 1 diabetes.³⁰ Several additional studies in people with type 2 diabetes have confirmed this finding. A cross sectional study of 548 people with type 2 diabetes from Brazil found that people with metabolic syndrome had increased prevalence of DPN (44% v 24%).³¹ Three smaller studies of people with type 2 diabetes in Australia (38% v 20%; n=148), Finland (16% v 6%; n=170), and the US (n=218) found that having the

metabolic syndrome increased the risk of DPN.³²⁻³⁴ One longitudinal study of 141 American Indian participants with longstanding type 2 diabetes found that a greater number of metabolic syndrome components was associated with a higher rate of neuropathy progression during a three year follow-up period.³⁵ Although fewer studies have assessed the association between the metabolic syndrome and DPN in type 1 diabetes, one large study from Australia (n=1914) found that having the metabolic syndrome increased the odds of DPN.³⁶ A second study of people with type 1 diabetes in Spain (10% v 4%; n=280) found that the metabolic syndrome increased the likelihood of DPN only when glycemic status was excluded.³⁷ Of note, the only study that included both people with type 1 diabetes and those with type 2 diabetes, the Metascreen study, found that metabolic syndrome had a larger effect in people with type 1 diabetes than type 2 diabetes.³⁰

Obesity

After hyperglycemia, obesity has emerged as the second leading metabolic risk factor for peripheral neuropathy in people with and without diabetes.³⁸ Multiple worldwide studies in people with diabetes living in Africa, Asia, Europe, and North America have found obesity to be associated with increased risk of DPN in people with type 2 diabetes.^{17 39-45}

The largest study evaluating associations between obesity metrics and DPN included 29 762 people with type 2 diabetes in the UK Biobank and found that larger body mass index (BMI) and waist circumference were associated with DPN.⁴² In addition, a secondary analysis of data from 7442 people with type 2 diabetes enrolled in the ACCORD trial found BMI and waist circumference to be associated with incident DPN.⁴⁵ Two longitudinal studies, the Anglo-Danish-Dutch Study of Intensive Treatment in People with Screen Detected Diabetes in Primary Care (ADDITION-Denmark; n=1256) and the Danish Centre for Strategic Research in Type 2 Diabetes (DD2; n=5249) study, also found that worsened obesity increased the likelihood of having DPN.^{43 44} Another notable study, the Japan Diabetes Complication and Prevention Prospective study found that higher BMI was associated with increased odds of DPN among 5451 people with diabetes.⁴¹ Furthermore, a study of 3021 people with type 2 diabetes from Qatar, Kuwait, and Saudi Arabia found higher BMI to be associated with increased odds of DPN.⁴⁰ Obesity has also been found to increase the likelihood of DPN in young people with type 2 diabetes. Specifically, the Treatment Options for Type 2 Diabetes in Adolescents and Youth (TODAY) study found that higher BMI increased the likelihood of DPN among 674 young people in the US with type 2 diabetes.¹⁷

Several large studies have also identified associations between obesity and DPN in people with type 1 diabetes. The largest study to date, based on data from the US T1D Exchange cohort, found that higher BMI was associated with DPN among

5936 people with type 1 diabetes.⁴⁶ Similarly, the SEARCH for diabetes in youth study found that among 1734 young people with type 1 diabetes, BMI and waist circumference were larger in those with DPN.¹⁶ Finally, the European Diabetes Prospective Complications (EURODIAB) study found higher BMI to be associated with incident DPN among 1172 people with type 1 diabetes.⁴⁷

Most studies assessing associations between obesity and DPN have assessed obesity by using simple anthropometric measurements, which are limited in their ability to distinguish between fat mass and fat-free mass. Only four studies have assessed the association between DPN and detailed obesity assessments, such as through bioelectrical impedance analysis or dual x ray absorptiometry and have found mixed associations with DPN. Specifically, studies in Korea (n=65) and Qatar (n=332) found measurements such as fat mass, fat percentage, and visceral adipose tissue volume to be associated with odds of DPN.^{48 49} By contrast, two studies in China (n=488 and n=2494) found that greater visceral fat area was associated with smaller odds of DPN.^{50 51} Considering this conflicting evidence, future studies are needed to determine whether detailed assessments of obesity can improve our understanding of DPN risk compared with simple anthropometric measures.

Other metabolic risk factors

Although severity and duration of hyperglycemia is the leading metabolic risk factor for DPN, along with obesity, evidence indicates that other individual metabolic syndrome risk factors, including triglycerides, high density lipoprotein (HDL) cholesterol, and hypertension also increase the risk of DPN.^{24 25}

A 2021 meta-analysis included data from 32 668 patients with diabetes from 39 studies and assessed associations of concentrations of triglycerides, HDL cholesterol, low density lipoprotein (LDL) cholesterol, and total cholesterol with DPN. In alignment with the individual risk factors included in the metabolic syndrome, the meta-analysis found that triglyceride and HDL cholesterol concentrations, but not LDL or total cholesterol concentrations, were associated with DPN.⁵² Notably, several recent large studies have confirmed the importance of triglycerides for DPN risk. A 2024 Danish study of 61 853 people with type 2 diabetes found that triglycerides, but not HDL or LDL cholesterol, were associated with increased risk of DPN.⁵³ Similarly, a large 2024 study of 13 315 adults with type 2 diabetes from China found triglycerides, but not HDL or LDL cholesterol, to be associated with DPN.⁵⁴ By contrast, in people with type 1 diabetes, the EURODIAB study found that total cholesterol, LDL cholesterol, and triglycerides each increased the odds of incident DPN.⁴⁷ Taken together, current evidence suggests that lipid profiles, particularly triglyceride concentrations, are important risk factors for DPN.⁵⁵

Multiple cross sectional studies have found that the presence of hypertension increases the risk of

DPN.⁵⁶ Cross sectional studies of adults with type 2 diabetes in China (n=14 908 and n=13 315) and across 14 countries in the International Prevalence and Treatment of Diabetes and Depression study found that hypertension increased the odds of DPN, even after comprehensive adjustment for other DPN risk factors.^{54 57 58} Furthermore, two cross sectional studies of Japanese adults with type 2 diabetes (n=9914 and n=5451) found higher systolic blood pressure to be associated with increased odds of DPN.^{41 59} By contrast, the ADDITION-Denmark and DD2 studies did not find hypertension or blood pressure to increase the likelihood of DPN.^{43 44} For people with type 1 diabetes, both the EURODIAB and SEARCH studies found that hypertension increased the likelihood of having DPN.^{16 47 58}

Discrepancies in the association between DPN and hypertension might be explained by differences in population characteristics, differences in DPN assessments, or lack of consistent statistical adjustment for other relevant risk factors. For example, use of antihypertensive drugs differs across these studies. Another explanation may be that the true effect size of hypertension on DPN may be small enough that some studies lack the sample size necessary to detect such an effect. Overall, a clear hierarchy of metabolic risk factors for neuropathy has been established, with diabetes leading the way, followed by obesity and then dyslipidemia and hypertension (table 1).

Social determinants of health

Social determinants of health (SDOH) are the social, economic, and political contexts that give rise to socioeconomic positions that result in exposures or vulnerabilities to conditions that compromise health.⁶⁰ Commonly assessed SDOH include household income, gender, education, and race/ethnicity. These factors act via a series of intermediary determinants that affect health and wellbeing, such as material circumstances (housing, neighborhood quality, and financial means), psychosocial circumstances (social support and living circumstances), and behavioral and biological factors (nutrition, genetics, and physical activity).⁶⁰

Multiple SDOH have been linked with the development of DPN in observational studies. Increased composite index scores have been associated with increased prevalence of DPN among adults with type 1 diabetes in Scotland and with increased use of drugs for painful DPN among adults with type 2 diabetes in England, even after control for other DPN risk factors such as duration of diabetes and age.^{61 62} In Romania, county level unemployment rates and gross domestic product per capita were associated with increased prevalence of DPN among patients with diabetes admitted to hospital.⁶³ SDOH assessed at the individual level have also been linked with DPN development. Among patients with type 1 diabetes enrolled in the US T1D Exchange Registry, those with DPN were more likely to be black, less likely to have a college education, and less likely to

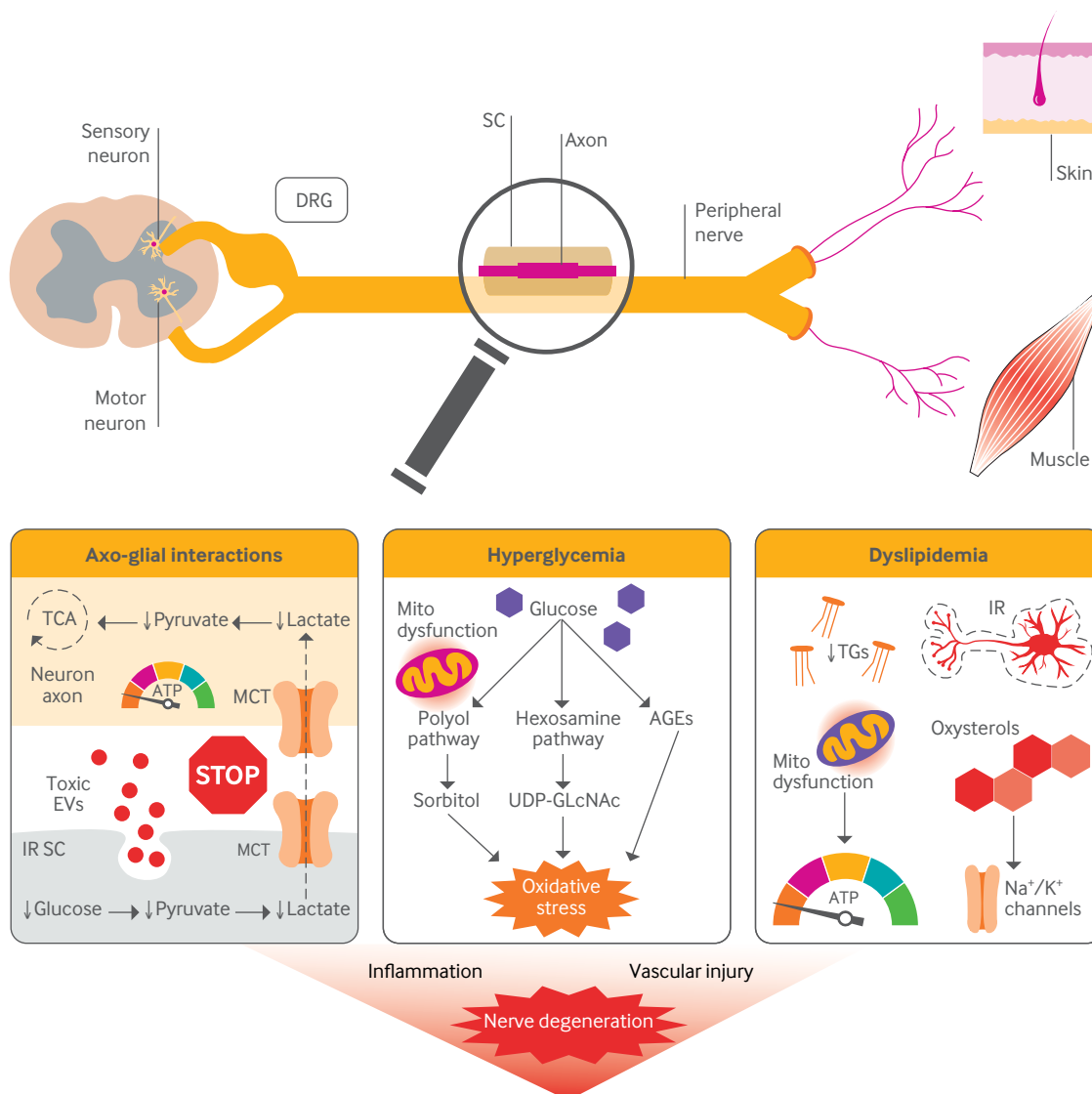


Fig 2 | Pathophysiology of diabetic peripheral neuropathy (DPN). DPN involves complex disruptions to sensory and motor neuron pathways that innervate the skin and muscle. Sensory neurons, with cell bodies in the dorsal root ganglia (DRG), relay information to the spinal cord, and motor neurons originate in the spinal cord ventral horn and transmit signals peripherally. Schwann cells (SCs), peripheral nerve glia, offer structural and metabolic support to both unmyelinated and myelinated axons. However, in DPN, axoglia interactions are impaired, featuring reduced lactate transport via monocarboxylate transporters (MCT), leading to ATP shortages as well as the release of pathological extracellular vesicles (EVs) from insulin resistant SCs (IR SC). Hyperglycemia exacerbates these effects through mitochondrial (mito) dysfunction and the activation of damaging metabolic pathways such as polyol and hexosamine, along with the accumulation of advanced glycation end products (AGEs). Concurrently, dyslipidemia contributes to nerve damage via increased triglycerides (TGs), insulin resistance (IR), the presence of oxysterols, and further mito impairment, leading to reduced ATP concentrations (indicated by the downward pointing ATP dial). These metabolic disturbances, in addition to inflammation and vascular injury, culminate in nerve degeneration. Original graphic created with BioRender.com

have private insurance coverage.⁴⁶ Together, these studies suggest that the prevalence of DPN may be greater in people in positions of socioeconomic disadvantage, although data remain limited.

A smaller but growing body of evidence supports an association between SDOH and DPN outcomes. In one US health system study of 144 564 individuals, adults with infected diabetic foot ulcers had greater levels of neighborhood disadvantage, as measured by the proportion of non-Hispanic black individuals, female headed households, and households with public assistance income or food stamps, than

adults with an uninfected diabetic foot ulcer.⁶⁴ Similarly, household income has been inversely associated with increased rates of amputation and mortality among 1362 patients with diabetic foot ulcers in a large South Korean claims database study.⁶⁵

The mechanisms by which SDOH lead to development of DPN have not been well established. These factors probably act directly and indirectly to influence health outcomes. Directly, SDOH such as income and education have been associated with increased risk of developing diabetes and disease

progression.⁶⁶ This relation may be indirectly mediated by access to quality diabetes care as people residing in neighborhoods with greater Area Deprivation Index scores were less likely to achieve optimal diabetes targets than those residing in neighborhoods with lower scores.^{67 68} However, in some cases, the mechanism of social determinants remains unclear. Higher levels of formal education have been associated with higher overall knowledge about DPN, but this does not necessarily translate to improved foot care behaviors, and interventions that increase knowledge about DPN do not necessarily lead to improved foot care practices.⁶⁹ A recent UK Biobank study of 24 297 participants also found that social isolation among patients with type 2 diabetes was associated with increased incidence of DPN.⁷⁰ Although this association may be mediated by unhealthy lifestyle factors, social isolation has also been shown to increase markers of systemic inflammation such as C reactive protein, which are also increased in DPN.^{71 72} Overall, this evidence suggests that interventions targeting SDOH will have to be multifaceted and target potential direct and indirect mechanisms of action on DPN prevalence and outcomes.

Pathophysiology

DPN involves complex pathological changes at the cellular level, particularly affecting neurons in the dorsal root ganglia (DRG) and their supportive glia, the Schwann cells. These components are essential for preserving nerve function and integrity, and their dysfunction contributes to disease progression. Evidence from experimental and clinical studies shows that these cellular disruptions are primarily driven by diabetes induced metabolic imbalances, such as hyperglycemia and dyslipidemia, which impair nerve metabolism, favor oxidative and pro-inflammatory conditions, and ultimately lead to bioenergetic failure (fig 2).⁷³

Neuronal and Schwann cell dynamics: importance of axoglial interactions

Early in the disease course, DRG neurons undergo changes affecting their axons, particularly at distal ends, as supported by evidence from animal models.⁷³ Notable changes include reduced production of neurofilament,⁷⁴ which is essential for axonal integrity and transport, and increased endoplasmic reticulum stress,⁷⁵ with subsequent disruption of protein folding. Alterations in key proteins that regulate axonal growth and plasticity, such as growth associated protein 43, have also been reported in small cross sectional studies and animal models.^{74 76 77} Beyond these molecular findings, recent large scale omics analyses uncovered significant mRNA and microRNA changes within murine and human peripheral nerves. These include enriched pathways involved in inflammation,⁷⁸⁻⁸⁰ mitochondrial dysfunction,^{81 82} and abnormal lipid metabolism,^{78 83} in both preclinical and clinical settings.

Although neuronal changes are important, they do not occur in isolation and cannot fully explain disease complexity. In addition to neurons, Schwann cells, historically known for structural support and myelination,⁸⁴ are now recognized for metabolically supporting axons, as shown *in vitro*, *in vivo*, and in knockout models.⁸⁵⁻⁸⁸ Under normal conditions, Schwann cells convert glucose into lactate,^{86 89} which is shuttled via monocarboxylate transporters, including MCT1, into axons.⁸⁸ There, lactate is converted to pyruvate to fuel axonal bioenergetics.⁹⁰ Schwann cells further promote axonal health by transferring extracellular vesicles to axons, which deliver microRNAs, metabolites, and proteins that modulate metabolic processes and enhance axonal growth and recovery.⁹¹

Metabolic stress impairs Schwann cell function, disrupting both myelination and metabolic communication with axons, thereby contributing to DPN. Recent single cell RNA sequencing data from the high fat diet model indicate that Schwann cells develop insulin resistance and a pro-inflammatory state, undermining their ability to support axons.⁹² These metabolic disturbances in Schwann cells and their effect on axonal health are further illustrated by Schwann cell specific knockout models, in which deletion of the energy sensor LKB1 or the lactate transporter MCT1 impairs energy homeostasis and reproduces essential features of DPN.^{86 88 93 94} Under hyperglycemic conditions, damaged Schwann cells exacerbate the situation by releasing pathologic extracellular vesicles that contain microRNAs such as miR-28 and miR-130a, which suppress axonal growth and accelerate nerve degeneration.⁹⁵ Although these findings are primarily derived from animal models, they call for confirmatory studies in humans to establish their clinical relevance.

Together, effective Schwann cell-axon interactions are vital for preserving axonal energy and integrity under normal conditions.^{86 89} Disruption leads to energy deficits and nerve degeneration, as supported by multiple animal studies in diabetic and non-diabetic models.^{88 93-95} This highlights the potential for targeting Schwann cells and their metabolic pathways to prevent DPN.

Hyperglycemia

Hyperglycemia was once regarded as the sole driver of DPN. Although we now recognize multiple contributing factors (for example, dyslipidemia, discussed below), hyperglycemia remains critically important. The streptozotocin induced type 1 diabetic rat model has been central to uncovering the pathogenic mechanisms of DPN, including polyol pathway activation, whereby excess glucose is converted to sorbitol. This process induces osmotic stress and Na⁺/K⁺-ATPase dysfunction, slowing nerve conduction.⁹⁶ The hexosamine pathway, as shown in similar animal models, diverts fructose-6-phosphate to uridine diphosphate *N*-acetylglucosamine, altering gene expression and promoting vascular injury and inflammation.⁹⁷

Advanced glycation end products (AGEs), formed from non-enzymatic reactions, accumulate in different parts of the nerve, including Schwann cells, axons, macrophages, and the vasculature, as shown by small cross sectional studies and experimental findings using the streptozotocin model and high glucose treatment in vitro.⁹⁸⁻¹⁰² On interacting with their receptors (RAGEs), AGEs activate signaling cascades that trigger inflammation, DNA degradation, decreased blood flow, and conduction deficits.¹⁰⁰⁻¹⁰³ Importantly, polyol, hexosamine, and AGE pathways all converge on oxidative stress as a common pathogenic mechanism.¹⁰⁴ Alongside these pathways, experimental studies on DRG neurons derived from type 1 diabetic animals show that hyperglycemia induces mitochondrial dysfunction, characterized by membrane depolarization and reduced ATP production.¹⁰⁵

Dyslipidemia

Mice fed a high fat diet develop obesity and dyslipidemia, but not hyperglycemia, as they progress to DPN.¹⁰⁶ In these models, lipidomic and transcriptomic profiling shows a significant accumulation of triglycerides, primarily comprising highly saturated acyl chains within peripheral nerves.⁸³ Such lipid build-up likely contributes to the insulin resistance observed in these nerves,¹⁰⁷ paralleling effects found in fat and muscle tissues. Research primarily conducted in animal and culture models has shown that insulin acts as a neurotrophic factor in healthy peripheral nerves, regulating key metabolic pathways such as AKT, AMPK, and PGC1 α .⁷³⁻¹⁰⁷⁻¹⁰⁹ In insulin resistant states, however, these processes are dysregulated, leading to inefficient nutrient and energy use.⁷³⁻¹⁰⁷⁻¹⁰⁹ High fat diet mice also show abnormal mitochondrial dynamics, characterized by increased mitochondrial size, decreased membrane potential, and altered intracellular calcium concentrations, which severely impair axonal signaling, maintenance, and regeneration.¹¹⁰ Additionally, distal sural nerve segments have lower basal ATP production and coupling efficiency than proximal DRG neurons, mirroring the distal-to-proximal axonal loss seen in DPN.¹¹¹

These in vivo findings are supported by in vitro studies in DRG neurons, which show that palmitate, a saturated fatty acid common in obesogenic environments, promotes axonal injury by reducing mitochondrial trafficking and triggering bioenergetic deficits.¹¹² Interestingly, these effects depend on fatty acid chain length and saturation, with medium chain laurate showing no effect and monounsaturated oleate offering neuroprotection.¹¹³⁻¹¹⁴

In addition to fatty acids, oxidized lipids, including cholesterol derived oxysterols and LDLs, adversely affect nerve health. Evidence, primarily from experimental studies, indicates that oxysterols can compromise myelin integrity and disrupt the function of sodium and potassium channels, potentially impairing nerve impulses.¹¹⁵ Moreover,

studies in animal and in vitro DPN models found that oxidized LDLs activate receptors such as LOX1,¹¹⁶ TLR4,¹¹⁷ and RAGE,¹⁰² initiating pro-inflammatory cascades and accumulation of reactive oxygen species,¹¹⁶ which further exacerbate DPN. Although these findings provide significant insights, further studies are needed to confirm their applicability in humans.

Genetics of diabetes and DPN

The development of both diabetes and its microvascular complications, including DPN, retinopathy, and nephropathy, have a strong genetic component.¹¹⁸⁻¹¹⁹ Genome-wide association studies (GWAS) in type 2 diabetes have shown that the genetic risk of diabetes for a given patient correlates with the overall burden of risk loci, which can be captured by a polygenic risk score.¹²⁰⁻¹²¹ Notably, this score is significantly associated not only with diabetes but also with development of microvascular complications.¹²⁰ The strongest type 2 diabetes risk gene identified to date, *TCF7L2* (odds ratio 3.3), which encodes a transcription factor involved in Wnt signaling, is also significantly associated with microvascular complications, including DPN (odds ratio 1.6).¹²¹ Overall, the polygenic risk score seems to be most predictive of the development of diabetic retinopathy, with more modest effects in diabetic nephropathy and DPN.¹²⁰ The latter result suggests that genetic susceptibility in DPN may be driven by distinct risk loci that operate independently from risk of diabetes alone. This is consistent with lower heritability of DPN compared with other vascular complications.

Familial studies in patients with type 1 diabetes suggested a 2.2-fold elevation in risk of DPN with an affected sibling,¹²² and GWAS have produced estimates of 11-15% for genetic risk of diabetic neuropathic pain and 6% for diabetic foot ulcers.¹²³⁻¹²⁴⁻¹²⁵ Comparison of genetic risk in DPN versus other diabetic complications suggests that susceptibility in DPN may be driven by at least some distinct risk loci that operate independently from risk of diabetes alone.

Many genomic studies have sought to identify such risk loci in DPN, but many studies have been candidate based studies or small GWAS with small sample sizes that limited their ability to produce results with genome-wide significance. Pooled analyses across several small studies suggest that specific variants in *ACE*, *MTHFR*, *GPx-1*, and *CAT* may be associated with risk of DPN,¹²⁶⁻¹²⁷ but these results need to be further replicated in larger populations.

To date, large scale GWAS in patients with DPN have been limited to five published studies (table 2). The first study, using a Scottish population of 7400 patients with diabetes (GoDARTS), found three genetic signals with a mild association with diabetic nerve pain, including a cluster of single nucleotide polymorphisms (SNPs) near *GFRA2* (lowest $P=1.77\times 10^{-7}$; odds ratio=0.67) that exerted a protective effect, a SNP near *ZSCAN20* and *TLR12P* (rs71647933; $P=3.84\times 10^{-7}$; odds

Table 1 | Key studies assessing associations between metabolic risk factors and diabetic peripheral neuropathy (DPN)

Author, year	Diabetes status	Country	Sample size	Study design	Metabolic risk factor	Findings
Anderson, 2018	T2D	Denmark	1256	Longitudinal cohort	Individual metabolic risk factors	Larger waist circumference (HR 1.14, 95% CI 1.05 to 1.24), higher BMI (1.14, 1.06 to 1.23), and lower HDL cholesterol (0.82, 0.69 to 0.99) were associated with increased hazard of DPN, after adjustment for age, sex, and randomization arm within ADDITION-Denmark clinical trial. HbA _{1c} , blood pressure, total cholesterol, and triglycerides were not associated with DPN
Bonadonna, 2006	Both T1D and T2D	Italy	8497 (7859 T2D, 638 T1D)	Cross sectional	Metabolic syndrome	Having metabolic syndrome was associated with increased odds of DPN in both T1D (OR 1.2) and T2D (OR 1.8), after adjustment for age, diabetes duration, HbA _{1c} , and LDL cholesterol
Cai, 2021	Both T1D and T2D	Meta-analysis	39 studies (32 668 people with T1D or T2D)	Meta-analysis	HDL, LDL, and total cholesterol and triglycerides	Pooled results from meta-analysis found that people with DPN had higher triglycerides (WMD 0.34, 95% CI 0.20 to 0.48) and HDL cholesterol (−0.05, −0.08 to −0.02), but no differences in LDL (0.00, −0.08 to 0.08) or total cholesterol (−0.03, −0.14 to 0.09), compared with those without DPN. In subset of studies that included estimates of OR, meta-analysis found that higher triglycerides (OR 1.36, 95% CI 1.20 to 1.54), lower HDL (0.85, 0.75 to 0.96) and higher LDL cholesterol (1.10, 1.01 to 1.19), but not total cholesterol (1.02, 1.00 to 1.04), were associated with DPN
Chen, 2023	T2D	China	13 315	Cross sectional	Individual metabolic risk factors	Hypertension (OR 1.08, 95% CI 1.00 to 1.17) and higher total cholesterol (1.11, 1.07 to 1.15), triglycerides (1.16, 1.14 to 1.18), and HbA _{1c} (1.03, 1.01 to 1.05), but not HDL or LDL cholesterol or BMI, were associated with increased odds of DPN, after adjustment for age, sex, other metabolic risk factors, and medication use
Christensen, 2020	T2D	Denmark	5249	Longitudinal cohort	Individual metabolic risk factors	Having large waist circumference (PR 1.67-2.45, depending on level), low HDL cholesterol (1.35, 95% CI 1.12 to 1.62), high HbA _{1c} (1.28-1.42, depending on level), and high triglycerides (1.36, 1.17 to 1.59) were associated with increased risk of DPN, after adjustment for age, sex, and diabetes duration. SBP, DBP, and total cholesterol were not associated with DPN
Costa, 2004	T2D	Brazil	548	Cross sectional	Metabolic syndrome	Having metabolic syndrome was associated with increased odds of DPN (OR 2.33, 95% CI 1.05 to 5.18), after adjustment for diabetes duration, HbA _{1c} , and total cholesterol
Huang, 2023	Both T1D and T2D	UK	29 762	Longitudinal cohort	Obesity (BMI and waist circumference)	Higher BMI (HR 1.27, 95% CI 1.20 to 1.34) and larger waist circumference (1.38, 1.30 to 1.45) were associated with increased risk of DPN, after adjustment for age, sex, race, education, blood pressure, HbA _{1c} , LDL cholesterol, and diabetes duration
Gababo, 2012	Both T1D and T2D	Ethiopia	528	Cross sectional	Individual risk factors	Having large waist circumference (>40 inches; OR 2.72, 95% CI 1.07 to 6.94), hypertension (2.36, 1.26 to 4.43), and high fasting blood glucose (2.55-3.88 depending on concentration) were associated with increased odds of DPN
Isomaa, 2001	T2D	Finland	170	Cross sectional	Metabolic syndrome	Prevalence of DPN was higher among those with metabolic syndrome (16% v. 6%)
Issar, 2021	T2D	Australia	148	Cross sectional	Metabolic syndrome	Prevalence of DPN was higher among those with metabolic syndrome (38% v 20%)
Jaiswal, 2017	T1D and T2D	US	1992 (1734 T1D, 258 T2D)	Longitudinal cohort	Individual metabolic risk factors	People with DPN had longer diabetes duration (for T1D and T2D), higher DBP (T1D only), higher BMI (T1D only), larger waist circumference (T1D only), higher LDL cholesterol (T1D only), higher triglycerides (T1D only), and lower HDL cholesterol (for T1D and T2D) compared with those without DPN. HbA _{1c} and SBP were not different between those with and without DPN. This analysis did not adjust for other risk factors
Kamiya, 2024	T2D	Japan	5451	Longitudinal cohort	Individual metabolic risk factors	Longer diabetes duration (OR 1.32, 95% CI 1.21 to 1.43), higher HbA _{1c} (1.15, 1.09 to 1.22), higher BMI (1.19, 1.09 to 1.30), and higher SBP (1.06, 1.01 to 1.10) were associated with increased odds of DPN, after adjustment for age, sex, metabolic risk factors, lifestyle factors, and medication use. Higher total cholesterol was associated with decreased DPN odds (0.98, 0.96 to 0.99)
Kristensen, 2024	T2D	Denmark	61 853	Longitudinal cohort	Triglycerides, HDL and LDL cholesterol	High triglycerides (HR 1.28, 95% CI 1.13 to 1.45), but not low HDL (1.09, 0.97 to 1.22) or high LDL cholesterol (1.03, 0.89 to 1.18), were associated with increased risk of DPN, after adjustment for age, sex, calendar year, smoking disorders, alcohol misuse, obesity, hypertension, and HbA _{1c}
Kristensen, 2023	T2D	Denmark	3397	Longitudinal cohort	β cell function and insulin sensitivity	Having hyperinsulinemia (high HOMA2-B and low HOMA2-S) was associated with increased prevalence of DPN (PR 1.35, 95% CI 1.15 to 1.57), after adjustment for age, sex, diabetes duration, diabetes medications, lifestyle factors, and metabolic syndrome risk factors
Lecumberri, 2022	T1D	Spain	280	Cross sectional	Metabolic syndrome	Having metabolic syndrome was associated with increased odds of DPN (OR 3.14, 95% CI 1.17 to 8.48), after adjustment for diabetes duration
Lee, 2021	T1D	Australia	1914	Cross sectional	Metabolic syndrome	Having metabolic syndrome was associated with increased odds of DPN (OR 1.5, 95% CI 1.1 to 1.9), after adjustment for age, diabetes duration, and HbA _{1c}
Lu, 2020	T2D	14 countries	2733	Cross sectional	Individual metabolic risk factors	Longer diabetes duration (OR 1.08, 95% CI 1.06 to 1.09), higher HbA _{1c} (1.11, 1.05 to 1.18), and having hypertension (1.58, 1.18 to 2.12), but not higher BMI (1.02, 1.00 to 1.04), were associated with DPN. Models were adjusted for age, sex, other demographics, smoking status, family history, exercise level, and other comorbidities

(Continued)

Table 1 | (Continued)

Author, year	Diabetes status	Country	Sample size	Study design	Metabolic risk factor	Findings
Mee-Kang, 2023	T2D	Korea	47 451	Longitudinal cohort	Obesity (BMI and waist circumference)	Those with DPN had larger waist circumference (85.5 v 85.3 cm) but lower BMI (24.8 v 25.1) than those without DPN. This analysis did not adjust for other risk factors
Mizokami-Stout, 2020	T1D	US	5936	Longitudinal cohort	Individual metabolic risk factors	Longer diabetes duration (OR 1.03, 1.02 to 1.04), higher HbA _{1c} (1.27, 1.17 to 1.37), higher BMI (1.03, 1.01 to 1.05), and higher triglycerides (1.03, 1.01 to 1.05) were associated with increased odds of DPN, after adjustment for age, sex, education, insurance status, height, statin use, and smoking status. HDL and LDL cholesterol were not associated with DPN
Oh, 2019	T2D	Korea	65	Cross sectional	Obesity (from BIA)	Higher obesity as measured by visceral fat area (1.03, 1.01 to 1.05), fat percentage (1.15, 1.03 to 1.28), and fat mass (1.13, 1.02 to 1.25), were associated with increased DPN odds after adjustment for age, sex, diabetes duration, and smoking status
Ponirakis, 2022	T2D	Qatar, Kuwait, and Saudi Arabia	3021	Cross sectional	Individual metabolic risk factors	Longer diabetes duration (OR 2.2-5.0, depending on length of diabetes, compared with those with diabetes duration of <10 years) and having poor glycemic control (1.6, 95% CI 1.2 to 2.0), obesity (2.0, 1.6 to 2.4), and hyperlipidemia (1.4, 1.1 to 1.8) were associated with increased odds of DPN after adjustment for age, sex, hyperlipidemia, hypertension, physical activity, smoking, and participant nationality
Ponirakis, 2025	T2D	Qatar	332	Cross sectional	Obesity (from DEXA)	Higher obesity, measured by visceral adipose tissue volume, was associated with DPN outcomes after adjustment for HbA _{1c} and hypertension
Reynolds, 2021	T2D	US	141	Longitudinal cohort	Metabolic syndrome	Higher number of metabolic syndrome risk factors was associated with increased rate of DPN progression (estimate: 0.05 (95% CI 0.01 to 0.10) per year per each metabolic syndrome factor), after adjustment for age, sex, height, and diabetes duration
Smith, 2013	T2D	US	218	Cross sectional	Metabolic syndrome	Presence of multiple metabolic syndrome components (≥3 of obesity, hypertension, low HDL cholesterol, and high triglycerides) was associated with increased odds of DPN (OR 3.0, 1.4 to 5.8)
Sun, 2024	T2D	China	488	Cross sectional	Obesity (from BIA)	Higher visceral fat area was associated with decreased odds of DPN (OR 0.98, 95% CI 0.98 to 0.99) after adjustment for diabetes duration and HbA _{1c}
Tesfaye, 2005	T1D	Europe	1172	Longitudinal cohort	Individual metabolic risk factors	Longer diabetes duration (OR 1.40, 95% CI 1.21 to 1.63), higher HbA _{1c} (1.48, 1.23 to 1.79), higher triglycerides (1.21, 1.02 to 1.44), higher BMI (1.27, 1.09 to 1.47), and hypertension (1.57, 1.03 to 2.39) were associated with increased odds of having DPN, after adjustment for smoking and albumin excretion rate. Total cholesterol was not associated with DPN
TODAY Study Group, 2021	T2D	US	674	Longitudinal cohort	Individual metabolic risk factors	Higher BMI (HR 1.28, 95% CI 1.15 to 1.43) and higher HbA _{1c} (1.26, 1.14 to 1.40), but not SBP, were associated with increased risk of DPN, after adjustment for age, sex, treatment group, C peptide, and hs-CRP
Wang, 2023	T2D	China	14 908	Cross sectional	Individual metabolic risk factors	Hypertension (1.36, 95% CI 1.25 to 1.48), longer diabetes duration (1.32-1.40, depending on length of diabetes duration), and higher HbA _{1c} (1.25-1.68, depending on level of HbA _{1c} , compared with HbA _{1c} <7%) were associated with increased odds of DPN. Models were adjusted for age, sex, education, and other diabetes complications. Higher BMI and lower HDL cholesterol were associated with decreased odds of DPN
Wu, 2022	T2D	China	2498	Cross sectional	Obesity (measured by BIA)	Higher visceral fat area was associated with lower DPN (OR 0.56 to 0.34, depending on quarter of visceral fat area compared with lowest quarter). Models were adjusted for age, sex, BMI, diabetes duration, other metabolic factors, comorbidities, and diabetes complications
Xing, 2021	T2D	China	1060	Cross sectional	Diabetes subgroups	DPN was more prevalent in SIRD (67%) than in SIDD (60%), MOD (48%), and MARD (14%)
Yokoyama, 2020	T2D	Japan	9914	Cross sectional	Individual metabolic risk factors	Longer diabetes duration and higher SBP, but not BMI, HbA _{1c} , HDL or LDL cholesterol, or triglycerides, were associated with increased odds of DPN, after adjustment for age, sex, alcohol, smoking, and medication use (ORs were not provided)
Zaharia, 2019	Both T1D and T2D	Germany	1105	Longitudinal cohort	Diabetes subgroups	DPN was more prevalent in SIDD (36%) than in SIRD (17%), MARD, 15%), MOD (11%), and SAID (5%)
Zhou, 2021	T2D	US and Canada	7442	Longitudinal cohort	Obesity (BMI and waist circumference)	Higher BMI (HR 1.30-1.77, depending on sex and obesity class) and larger waist circumference (1.15-1.42, depending on sex and waist circumference quarter) were associated with incident DPN, after adjustment for age, race, medications, height, lifestyle factors, diabetes duration, HbA _{1c} , HDL cholesterol, hypertension, depression, and other microvascular diabetes complications

ADDITION=Anglo-Danish-Dutch Study of Intensive Treatment in People with Screen Detected Diabetes in Primary Care; BIA=bioelectrical impedance analysis; BMI=body mass index; CI=confidence interval; DBP=diastolic blood pressure; DEXA=dual energy x ray absorptiometry; HbA_{1c}=hemoglobin A_{1c}; HDL=high density lipoprotein; HOMA2-B=homeostatic model assessment - β cell function; HOMA2-S=homeostatic model assessment - insulin sensitivity; HR=hazard ratio; hs-CRP=high sensitivity C reactive protein; LDL=low density lipoprotein; MARD=mild age related diabetes; MOD=mild obesity related diabetes; OR=odds ratio; PR=prevalence ratio; SBP=systolic blood pressure; SAID=severe autoimmune diabetes; SIDD=severe insulin deficient diabetes; SIRD=severe insulin resistant diabetes; T1D=type 1 diabetes; T2D=type 2 diabetes; WMD=weighted mean difference.

ratio=1.66), and a SNP near *HMGB1P46* (rs6986153; lowest $P=8.02 \times 10^{-7}$; odds ratio=1.67).^{123 124} GFRA2 is a primarily neuronally expressed neurotrophic factor receptor that has been implicated in nociception in mice.¹²⁸ The function of ZSCAN20 is unknown, and *HMGB1P46* and *TLR12P* are pseudogenes with no known protein products.¹²⁴ Another analysis of this same patient cohort identified a SNP in *MAPK14* (rs80028505; $P=2.45 \times 10^{-8}$; odds ratio=1.71) associated with foot ulcers in patients with DPN.¹²⁵ *MAPK14* encodes one of the four p38 mitogen activated protein kinases that function to regulate transcription in response to varied extracellular stressors.¹²⁹

Another genome-wide association study analyzed two cohorts of controls and several thousand patients with type 2 diabetes (ACCORD and BARI 2D) and identified a locus near the voltage gated sodium channel gene *SCN2A* that exerted a strong protective effect against DPN (rs1341783; $P=1.9 \times 10^{-9}$; odds ratio=0.64).¹³⁰

Another genome-wide association study using the DOLORisk cohort of 2908 patients with painful and painless neuropathy of various causes found genome-wide significance for SNPs in *KCNT2* (rs114159097 $P=3.55 \times 10^{-8}$; $\beta=0.51$) and *NOS1AP* (rs10919166; $P=1.07 \times 10^{-9}$; $\beta=-0.77$) as contributors to neuropathic pain intensity in diabetic neuropathy.¹³¹ These genes encode a sodium activated potassium channel expressed in DRG neurons and a nitric oxide synthase adaptor protein, both of which have been previously implicated in animal studies of pain.^{132 133} Using gene burden analysis, this study also identified genetic variants in *OPRM1* (encoding the μ opioid receptor) and *SCN9A* (encoding a voltage gated sodium channel) that were more common in painful diabetic neuropathy. These results are intriguing given the known role for pathogenic variants in *SCN9A* in hereditary erythromelalgia and congenital insensitivity to pain and the role of opioid receptors in pain modulation.

Finally, a much larger genome-wide association study that included more than 200 000 cases (~40 000 with DPN) and 1 100 000 controls identified a variant near *NRP2* (rs78977169; $P=3.4 \times 10^{-8}$) as a significant risk factor for DPN. This gene encodes neuropilin 2, a receptor involved in axon guidance and angiogenesis.¹²⁰

Several studies have also investigated potential associations between epigenetic changes and the presence of DPN and neuropathic pain. One small study identified altered DNA methylation signatures from sural nerves of patients with stable/improving DPN ($n=6$) versus those with worsening DPN ($n=6$).¹³⁴ This work highlighted alterations in several biological processes, perhaps most notably MAPK signaling. A larger study comparing methylation in whole blood from two cohorts of patients with painful DPN ($n=99$ in total) and painless DPN ($n=132$ in total) found changes in nine candidate genes, although the magnitude of the changes was small.¹³⁵ These studies do not establish a causal role for epigenetic changes in DPN or neuropathic pain,

but they represent an important direction for further research.

Together, these results have begun to identify specific genes and common pathways that might confer susceptibility to DPN. However, insights into the genetic determinants of DPN have failed to keep pace with discoveries for other diabetic complications. This gap in understanding likely reflects the low emphasis on DPN in GWAS to date and fundamental difficulties in DPN research, including challenges and inconsistencies in methods to define and phenotype patients with DPN and the complex nature of medical and environmental contributors to DPN. However, with the increasing development and growth of large scale clinical-genomic databases, such as the Million Veteran Program, UK BioBank, and All of Us,¹³⁶⁻¹³⁸ opportunities exist to overcome these challenges and further disentangle the genetic contributions to DPN.

Diagnosis

History and examination

The diagnosis of DPN primarily relies on clinical history and neurologic examination.¹³⁹ The most common symptoms are numbness, tingling, and/or pain starting in the toes and ascending the legs in a stocking-glove distribution.

Neurologic examination often reveals length dependent changes in pinprick/temperature sensation (small fiber) and/or vibration/proprioception sensation (large fiber). Large fiber sensation can be assessed using a 128 Hz tuning fork on the great toe. Examination with a 10 g monofilament can also be used to evaluate for loss of protective sensation and risk of ulceration; however, monofilament does not reliably detect early DPN. Reflexes in DPN are often decreased or absent in a length dependent fashion. Weakness is usually a later finding in patients with severe DPN, with weakness starting in toe extension. Finally, a careful examination of the foot architecture and skin is important in identifying mild deformities, calluses, or skin breakdown that can increase the risk of ulcers.

Past medical history is important to identify potential causes of neuropathy other than diabetes. Importantly, diabetes is by far the most common cause of neuropathy, accounting for up to half of all cases of neuropathy.¹⁴⁰ Family history is important to inform the chance of an inherited cause. An extensive alcohol history should be obtained given that alcoholic neuropathy is one of the most common causes of neuropathy, with almost half of chronic alcohol misusers affected.¹⁴¹ Atypical features on clinical history and neurologic examination, such as acute/subacute onset, non-length dependence, motor predominance, and asymmetry, should prompt considerations of alternative diagnoses and potentially other causes.¹⁴²

Investigations

The laboratory evaluation of DPN should include a complete blood count, comprehensive metabolic

panel, vitamin B₁₂ concentration, and serum protein electrophoresis with immunofixation.¹⁴³ Importantly, diabetes is commonly identified at the time of diagnosis of neuropathy with testing such as hemoglobin A_{1c} and/or glucose tolerance testing.¹⁴⁴

More extensive laboratory testing is not needed unless atypical features are present as discussed above.¹⁴⁵ Electrodiagnostic studies are usually not needed for a diagnosis of DPN, as they rarely change the diagnosis or management.¹³⁹ Therefore, these tests should be reserved for patients with atypical features and/or where doubt exists about the relative contributions of multiple peripheral nervous system conditions. As a result, the American Diabetes Association's position statement recommends against routine use of electrodiagnostic testing for the evaluation of DPN.¹

Similarly, magnetic resonance imaging (MRI) of the brain and spinal cord is not needed in the evaluation of patients with suspected DPN, as it rarely changes the diagnosis or management.¹³⁹ Choosing Wisely guidelines recommend against ordering MRI for the evaluation of neuropathy.¹⁴⁶ Electrodiagnostic testing and MRI are the main drivers of cost in the evaluation of DPN, so cost effective approaches to diagnosis of DPN should reserve these tests for atypical cases.¹⁴⁷ Unfortunately, despite the guidelines, electrodiagnostic testing and use of MRI have not changed from 2007 to 2019 despite frequent use.¹⁴⁸

Similarly to electrodiagnostic studies for large fiber involvement, skin biopsies are the best confirmatory test for small fiber involvement but are rarely needed to make a clinical diagnosis of DPN.¹⁴⁹ Corneal confocal microscopy also evaluates small fiber involvement but is not needed for the diagnosis of DPN.¹⁵⁰⁻¹⁵² Overall, DPN is a clinical diagnosis that requires simple blood tests, with further testing reserved for individuals with atypical features. Improvements in diagnostic practices are needed, but which provider and system level interventions would optimize practice is uncertain.

Treatment

Disease modifying treatment

Disease modifying treatments for DPN are limited. Control of hyperglycemia is the main intervention available. Unfortunately, the effect of enhanced hyperglycemia control on prevention of DPN is limited in patients with type 2 diabetes, with a small effect observed (annualized risk difference of neuropathy -0.58, 95% confidence interval (CI) -1.17 to 0.01) in a meta-analysis (n=6669) that did not quite meet statistical significance.¹⁵³ By contrast, hyperglycemia control had a much more robust effect on reducing DPN in patients with type 1 diabetes (annualized risk difference of neuropathy -1.84, 95% CI -2.56 to -1.11; n=1228). Dietary weight loss in conjunction with exercise counseling had a modest effect on neuropathy outcomes in a large and long randomized controlled trial (n=5145 over 9-11 years; mean Michigan Neuropathy

Screening Instrument (MNSI) questionnaire score in intervention group 2.21 (standard deviation (SD) 0.03) versus 2.35 (0.03) in education control group; P<0.001).¹⁵⁴

An observational study also reported comparable results (n=72 over 2 years; mean MNSI questionnaire score improved by 0.6 (SD 1.4); P<0.01).¹⁵⁵ Similarly, observational studies of surgical weight loss show improvements in neuropathy outcomes (n=79 over 2 years; intraepidermal nerve fiber density (IENFD) thigh improvement of 3.4 (SD 7.8); P<0.01), but randomized trials are not available and most studies are small.³⁸ Many small exercise studies have also been conducted, with most showing improvements in neuropathy outcomes including nerve fiber density on skin biopsy.¹⁵⁶ However, a large definitive randomized study of exercise interventions has not been published to date. Taken together, disease modifying treatment includes a focus on hyperglycemia, diet, and exercise, which can all be managed by primary care providers.

Neuropathic pain treatment

Drugs

Given limitations with disease modifying therapies, most of the treatment of DPN focuses on treatment of neuropathic pain. Interestingly, a significant proportion of patients with DPN do not experience pain.^{8 157} On the other hand, those with neuropathic pain experience worse quality of life, depression, anxiety, and sleep, making this a critical aspect of treatment.⁸ Several guidelines have recommended tricyclic antidepressants, serotonin norepinephrine reuptake inhibitors (SNRIs), and gabapentinoids for the treatment of painful DPN.¹⁵⁸⁻¹⁶²

More recently, the American Academy of Neurology guideline and meta-analysis confirmed these findings and added sodium channel blockers as an effective class.¹⁶³ Unfortunately, the effect sizes for these drugs are relatively modest (standardized mean differences 0.44-0.95), and all have potential side effects. One comparative effectiveness study, OPTION-DM (n=130), showed that amitriptyline (tricyclic antidepressant), duloxetine (SNRI), and pregabalin (gabapentinoid) had the same efficacy (mean difference -0.01 or less on pain numerical rating scale) and that combining drugs from different classes has additional benefit over monotherapy (mean reduction of 1.0 (SD 1.3) v 0.2 (1.5) on pain numerical rating scale).¹⁶⁴ Another pragmatic comparative effectiveness study in idiopathic neuropathy (n=402) showed that nortriptyline and duloxetine (probabilities of highest utility 0.52 and 0.43) were superior to pregabalin and mexiletine (probabilities of highest utility 0.05 and 0.00) for the treatment of neuropathic pain.¹⁶⁵ Importantly, pregabalin had problems with insurance approval in the US and mexiletine with tolerability. Therefore, the efficacies of tricyclic antidepressants, SNRIs, gabapentinoids, and sodium channel blockers are likely similar given the results of a meta-analysis based on a considerable number of large, randomized trials and the OPTION-DM trial.

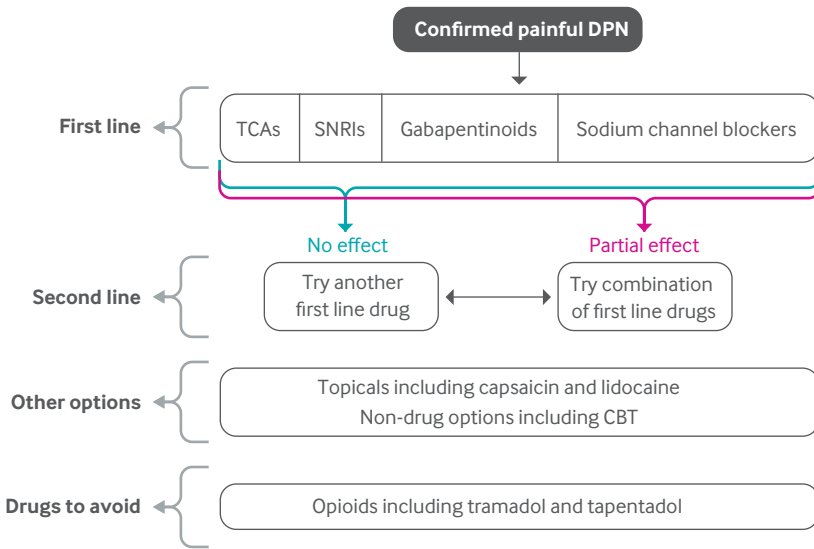


Fig 3 | Treatment algorithm for painful diabetic peripheral neuropathy (DPN) based on the 2022 American Academy of Neurology guideline and meta-analysis. CBT=cognitive behavioral therapy; SNRI=serotonin-norepinephrine reuptake inhibitor; TCA=tricyclic antidepressant

Given the similar efficacy, other factors should be taken into account when choosing which drug to start, including medical comorbidities and cost (fig 3).¹⁶⁶ Total costs, and in the US out-of-pocket costs, of drugs for neuropathic pain can be quite high, especially when they are not available as generics.^{167 168} Fortunately, generic options are available in all four drug classes. Other pain treatment options include topical options such as capsaicin cream and lidocaine patches.^{160 163} Systematic reviews identified three randomized controlled trials showing efficacy of capsaicin in DPN,^{159 163} including two using 0.075% cream and one using an 8% patch.¹⁶⁹⁻¹⁷¹ Given that the 0.075% capsaicin cream is available over the counter, is cheaper, and does not require medical personnel to apply it, this formulation is preferred to the 8% patch. Similarly, lidocaine patches are available in 4% and 5% versions, with the 4% version favored because it is available over the counter and cheaper.

Non-drug treatment

Among non-drug approaches for DPN, cognitive behavioral therapy (CBT) is the best studied.¹⁷²⁻¹⁷⁴ A meta-analysis found that CBT (n=4 trials) and mindfulness (n=3 trials) had similar reductions in painful DPN, but only CBT had a statistically significant effect.¹⁷⁵ Similarly, a meta-analysis showed that acupuncture (n=12 trials) significantly reduces painful DPN.¹⁷⁶

All the above options are preferred to opioids given the long term risks associated with these drugs, including misuse, addiction, overdose, fractures, and myocardial infarction.¹⁷⁷ Although combination SNRI-opioid drugs such as tramadol and tapentadol have shown efficacy in patients with painful DPN, this class of medication also has the same downsides as other opioids, including increased mortality; therefore, these drugs should largely be avoided.¹⁷⁸ A non-masked, randomized study of spinal cord stimulation showed improvements in painful DPN, but these results should be interpreted cautiously given that non-masked pain studies are hard to interpret and sham controlled studies for radicular pain have shown no benefit.^{179 180}

Taken together, painful DPN can be treated with four different drug classes, two topicals, and CBT. Opioids are commonly used but should not be used for chronic, non-cancer pain given the large number of adverse effects.^{181 182} Unfortunately, pain response often remains insufficient, and the polypharmacy often needed for the treatment of DPN can be problematic. The investigation of new pain and disease modifying treatments therefore remains critical.^{160 183 184} Importantly, the treatment of painful DPN does not require specialist involvement, as all the interventions are commonly used by primary care physicians.

Emerging treatments

Newer drugs for diabetes, glucagon-like peptide-1 (GLP-1) receptor agonists, sodium-glucose cotransporter 2 (SGLT2) inhibitors, and dipeptidyl peptidase 4 (DPP4) inhibitors, are now available, but whether they affect neuropathy outcomes remains to be determined. Glucagon-like peptide-1 receptor agonists and dipeptidyl peptidase 4 inhibitors have been best studied to date, but only a few, small,

Study design (PMID)	Sample size	Intervention/control	Outcomes	Results
GWAS to detect risk for pain in DPN (24974787)	572 painful DPN cases	2491 controls with diabetes	SNP near <i>GFR2</i>	OR=0.6; P=1.77×10 ⁻⁷
GWAS to detect risk for pain in DPN (26629533)	961 painful DPN cases	3260 controls with diabetes	SNP near <i>ZSCAN20-TLR12P</i>	OR=1.66; P=3.84×10 ⁻⁷
GWAS to detect risk for diabetic foot ulcers (28672053)	699 diabetic foot ulcer cases	2695 controls	SNP in <i>MAPK14</i>	OR=1.71; P=2.45×10 ⁻⁸
GWAS to detect risk of DPN (31127053)	2 cohorts: 4384 DPN cases and 791 DPN cases	2 cohorts: 784 controls and 158 controls (diabetes without neuropathy)	SNP near <i>SCN2A</i>	OR=0.64; P=1.9×10 ⁻⁹
GWAS to detect risk of neuropathic pain in DPN (39471050)	1541 patients with diabetic neuropathy	-	SNPs in <i>KCNT2</i> and <i>NOS1AP</i>	β=0.51; P=3.55×10 ⁻⁸ and β=-0.77; P=1.07×10 ⁻⁹
GWAS to detect risk for vascular outcomes in diabetes (32541925)	228 499 patients with diabetes, 40 475 with neuropathy	1 178 783 controls	SNP near <i>NRP2</i>	P=3.4×10 ⁻⁸

OR=odds ratio; SNP=single nucleotide polymorphism.

randomized trials have been completed with mixed results.¹⁸⁵⁻¹⁸⁷ Large, well conducted trials are needed to determine the effects of these drug classes on DPN outcomes. Similarly, a small, randomized trial of topical oxybutynin improved nerve fiber density in patients with type 2 diabetes, but larger trials with clinical outcomes are needed before recommendations for clinical use can be made.¹⁸⁸ For painful DPN, a phase 2a study of LX9211, an adaptor protein-2-associated kinase 1 showed promising results for the low dose but not the high dose group (n=319).¹⁸⁹ A phase 2b study was recently completed for LX9211 but has not been published to date.

Clinical trial readiness

Multiple clinical outcome assessments and patient reported outcome measures have been developed for the assessment of DPN diagnosis, severity, and pain (table 3). Among these are the commonly used MNSI questionnaire and examination scores,¹⁹⁰⁻¹⁹² Toronto Clinical Scoring System,¹⁹³ and modified Toronto Clinical Neuropathy Score (mTCNS).¹⁹⁴ A study comparing seven clinical scores in 81 participants with impaired glucose tolerance and neuropathy found that all of the scores effectively identified participants with early DPN, with the mTCNS showing the highest specificity and sensitivity (area under the receiver operating characteristic curve 0.998, 95% CI 0.93 to 0.99).¹⁹⁵ Although the MNSI was initially developed as a screening instrument rather than a measure of progression, a recent large prospective study (n=389) confirmed five year increases in both the MNSI examination and mTCNS scores in participants with probable and confirmed DPN.¹⁹⁶ This study also showed agreement between the clinical assessments, nerve conduction studies (NCS), and IENFD in patients with established DPN.¹⁹⁶ In addition to disease specific outcomes, clinical trials in neuromuscular diseases have increasingly incorporated the Patient Global Impression of Change, which was shown to correlate with established measures of pain intensity

in a cohort of participants with neuropathic pain (n=495).¹⁹⁷

An important challenge in trial design for DPN is that the intrinsically slow progression of neuropathy necessitates larger sample sizes and longer study durations to detect response to treatment. Furthermore, progression of DPN is not always linear, with periods of stabilization and even regression throughout its clinical course.¹⁹⁸⁻¹⁹⁹ A recent natural history study of 389 patients with newly diagnosed type 2 diabetes, 151 of whom were followed over five years, confirmed regression of neuropathy in a subset of patients as evidenced by both NCS and IENFD (5.9% and 18.6% participants, respectfully).¹⁹⁶ Furthermore, whereas most DPN affects both small and large nerve fibers (that is, Aβ, Aδ, and C fibers), a minority of patients manifest with small or large fiber predominant phenotypes requiring the use of differential outcomes.²⁰⁰⁻²⁰¹

Clinical trials in other slowly progressive neuropathies have underscored the importance of incorporating surrogate biomarkers that can be more responsive to change than bedside assessments.²⁰² In DPN, NCS have historically been used in clinical trials; however, their research utility has been questioned owing to their inability to detect the early small fiber involvement frequently seen in DPN. IENFD on skin biopsy is helpful in evaluating small fiber injury and has been found to be safe²⁰³; however, its performance as a treatment responsive biomarker has yet to be tested in large clinical trials. Similarly, corneal confocal microscopy has also been studied as a potential treatment responsive biomarker.¹⁵⁰⁻²⁰⁴⁻²⁰⁵ Continued head-to-head comparisons between IENFD and corneal confocal microscopy are needed to determine the best biomarker for small fiber injury. Transcutaneous electrical stimulation approaches have also been tested as DPN biomarkers, but they require specialized training and equipment.²⁰⁶ Finally, multiple studies have highlighted the potential use of MRI and ultrasonography in diagnosis of DPN, although most of these studies have been cross sectional.²⁰⁷⁻²¹⁷

Table 3 | Clinical outcome assessments used in diabetic peripheral neuropathy (DPN)

Assessments	Summary of assessments	PMIDs
Michigan Neuropathy Screening Instrument (MNSI)	Self-administered questionnaire and lower extremity examination	7821168, 16150538, 22417277
Toronto Clinical Neuropathy Score (TCNS)	Symptom questionnaire and sensory and reflex examination	12401755
Modified Toronto Clinical Neuropathy Score (mTCNS)	Modified to optimally assess early DPN. Symptom questionnaire and sensory examination	19317818
Utah Early Neuropathy Scale (UENS)	Motor, sensory, and reflex examination in early DPN	18844788
Neuropathy Disability Score (NDS)	Neurological examination	10840990
Neuropathy Impairment Score of the lower limb (NIS-LL)	Neurological examination focusing on lower limbs	10023123
Michigan Diabetic Neuropathy Score (MDNS)	Quantitative neurological examination and nerve conduction studies	7821168
Total Neuropathy Score-clinical (TNS-C)	Neurological examination focusing on lower limbs	10563609
Neuropathic pain and quality of life assessments		
Douleur Neuropathique en 4 questions	Sensory symptoms and examination	22023377, 15733628
Brief Pain Inventory for patients with painful diabetic peripheral neuropathy (BPI-DPN)	Patient reported assessment of pain severity and interference	15857744
PainDETECT	Patient reported assessment of pain gradation and pain course	17022849
Neuropathy and foot ulcer specific quality of life instrument (NeuroQoL)	Patient reported impact of DPN and foot ulcers on quality of life	12941717
Norfolk Quality of Life-Diabetic Neuropathy (QOL-DN)	Patient report of symptoms, and impact of activities of daily living	15929681

Among plasma biomarkers, the best studied in neuropathy to date is neurofilament light chain (NFL), a cytoskeletal protein released in response to axonal injury. Plasma NFL elevations have previously effectively heralded treatment response in several autoimmune and inherited neuropathies,²¹⁸⁻²²⁰

and several studies have recently evaluated the potential utility of NFL in DPN.²²¹⁻²²⁵ An evaluation of 178 participants of the ADDITION-Denmark study identified a greater longitudinal change in NFL in those with ultimate DPN and higher yearly increases associated with higher risk of DPN (odds

Table 4 | Studies of prevalence of diabetic peripheral neuropathy

Author, year	Country	Population	DPN measure	Outcome	Strengths and limitations	PMID/DOI
Adeniyi, 2015	Nigeria	264 T2D from two clinics	MNSI-Q, MNSI-E, monofilament	Prevalence: 26.1%	Limitations: clinic based recruitment	https://journals.co.za/doi/abs/10.1080/16089677.2015.1056482
Andersen, 2018	Denmark	1256 adults with T2D	MNSI-Q	Incidence: 10% over 13 years	Strength: primary care patients; limitation: average participant aged 60, so may not apply to younger cohorts	29487078
Arellano-Longinos, 2017	Mexico	106 adults with T2DM	MNSI-Q and MNSI-E	Prevalence: 81.1%	Strength: included examination; Limitations: small sample from single center, predominantly male	https://doi.org/10.22201/facmed.14058871p.2018.1.62907
Bello, 2019	Nigeria	175 T2D; hospital based study	MNSI-Q and MNSI-E	Prevalence: 41.7%	Strength: other risk factors for neuropathy were assessed	31187753
Cardinez, 2018	Canada	361 T1D aged >50 years	MNSI-Q	Prevalence: 42.7%	-	29929836
Christensen, 2020	Denmark	5249 T2D	MNSI-Q	Prevalence: 17.9%	Strength: well phenotyped patient cohort	32295810
Dabelea, 2017	US	1746 T1D and 272 T2D	MNSI-E	T1D prevalence: 8.5%; T2D prevalence: 17.7%	Strength: younger patient population (T1D mean age 17.9 years, T2D 22.1 years)	28245334
Del Brutto, 2016	Ecuador	110 adults with T2D	MNSI-Q and MNSI-E	Prevalence: 59%	Strength: population based recruitment in community	27718386
D'Souza, 2015	India	208 T2D >5 years	MNSI-Q and MNSI-E	MNSI-Q prevalence: 18.3%; MNSI-E prevalence: 32.2%	Limitations: hospital based cohort, other etiologies of DPN not assessed	https://journals.sagepub.com/doi/full/10.4081/jphr.2015.450
Gylfadottir, 2020	Denmark	5514 adults with T2D	MNSI-Q	Prevalence: 18%	Strength: nationwide cohort study; limitations: no physical examination, cohort enrolls patients from hospital outpatient clinics, meaning that they may have greater severity of disease, no details on other diabetic comorbidities	31693539
Ibarra, 2012	Mexico	348 T2D	MNSI-Q and MNSI-E	Prevalence: 69%	Limitations recruited from 1 primary care clinic, so may not represent community	23354633
Jember, 2017	Ethiopia	368 T1D and T2D from one hospital	MNSI-Q and MNSI-E	Prevalence: 52.2%	Limitation: single hospital study	28396852
Jeyam, 2020	Scotland	5558 T1D	MNSI-Q	Prevalence: 13%	Strength: population representative sample	31974100
Lee, 2014	Taiwan	404 T2D recruited from clinic registration files	MNSI-E	Prevalence: 34.5%	Limitation: participants with amputations were excluded	25421066
Maiga, 2020	Mali	252 adults recruited from diabetes care clinics	MNSI-I and MNSI-Q	Prevalence: 69.8%	Strength: examination and questionnaire completed; limitation: recruited from diabetes care centers, so data may have limited external validity	33166296
Mizokami-Stout, 2020	US	5936 T1D	MNSI-Q	Prevalence: 11%	Strength: large, well characterized patient cohort	32029635
Pai, 2018	Taiwan	2837 adults with T2D; hospital based study	MNSI-Q and MNSI-E	Prevalence: 21.3%	Limitation: hospital based cohort, so participants may have more severe disease	29042249
Pop-Busui, 2009	US	2368 T2D	MNSI-E	Prevalence: 51%	Limitation: cohort has coronary artery disease, so results may not be applicable to all T2D	19335534
Rivas Acuna, 2017	Mexico	198 patients admitted to hospital with T2D	MNSI-Q, MNSI-E, monofilament; diapason for vibration	Prevalence: 65.7%	Strength: additional testing for confirmation also completed; limitation: hospital based cohort	https://revistas.udes.edu.co/cuidarte/article/view/348/751
TODAY study group, 2021	US	674 young people with T2D	MNSI-Q and MNSI-E	Prevalence: 34.9%	-	34716210
Ugoya, 2006	Nigeria	120 adults with diabetes; 60 age matched controls without diabetes from one hospital	MNSI-Q	Prevalence: 75%	Limitation: MNSI-E tested, but diagnostic cut-offs not used	17128685
Wu, 2007	France	1023 from community of T1D and T2D	MNSI-Q	Prevalence: 11%	Strength: community based random sample; limitation: included only individuals with landlines, and phone calls were made during day, so most participants were women and retired	17637204

MNSI-E=Michigan Neuropathy Screening Instrument- Examination; MNSI-Q=Michigan Neuropathy Screening Instrument-Questionnaire; T1D=type 1 diabetes; T2D=type 2 diabetes.

ratio 1.36, 95% CI 1.08 to 1.71).²²⁴ Similarly, a 12 year retrospective case-control study from the TODAY cohort (n=100) showed more rapid increases in NFL in those with ultimate DPN (P=0.05). By contrast, a recent study of 60 adolescents with type 1 diabetes found no difference in NFL between those with and without DPN.²²⁶ Given that NFL also reflects injury to central nerve axons, new biomarkers with higher specificity for peripheral nerve axons and Schwann cells are needed.^{227 228}

Despite notable efforts to ensure trial readiness for DPN, no unified definition exists for DPN in the research setting, and the use of varied instruments continues to complicate the translation of preclinical insights to treatments that can benefit patients.^{184 229} This challenge will need to be tackled through collaboration across the DPN research community. Such efforts are exemplified by a recent gap analysis conducted by expert clinicians and researchers in the DPN field that showed a clear consensus on the importance of using patient reported outcome measures in clinical trials, the utility of IENFD as a research outcome measure for DPN, and the need for new outcomes better suited for the timeframe of clinical trials.²³⁰

Guidelines

The American Diabetes Association (ADA) publishes an annual "Standards of Care in Diabetes" guideline that provides recommendation on DPN screening and treatment. In 2026 the ADA recommends DPN screening at diagnosis of type 2 diabetes and five years after diagnosis of type 1 diabetes and annually thereafter in both groups.²³¹ Specifically, the screening assessment should include a test for small fiber neuropathy (pinprick or temperature sensation) and a test for large fiber neuropathy (vibration). In addition, 10 g monofilament testing is suggested to evaluate for risk of ulcer. For treatment, the ADA recommends optimizing glucose management, weight, blood pressure, and lipids.²³¹

Most DPN guidelines pertain to the treatment of painful DPN. Fortunately, a consensus exists in several guidelines that gabapentinoids, SNRIs, and tricyclic antidepressants are effective at reducing DPN pain.¹⁵⁸⁻¹⁶³ Differences in the guidelines mainly focus on the role of opioids, sodium channel blockers, and topical medications. Three guidelines discuss supportive evidence for opioids in DPN with two providing recommendations for opioids as second or third line options.^{158 160 162} By contrast, the American Academy of Neurology and ADA guidelines explicitly recommend against the use of opioids, including tramadol and tapentadol (SNRI/opioids) for painful DPN.^{163 231} Furthermore, three guidelines provided evidence supporting two specific sodium channel blockers (carbamazepine and oxcarbamazepine), with another reporting that evidence for the efficacy of multiple sodium channel blockers was inconclusive.¹⁵⁸⁻¹⁶² However, the first guideline evaluating for class effects, rather than the effects of specific drugs, found that sodium channel

blockers have similar efficacy to gabapentinoids, SNRIs, and tricyclic antidepressants.¹⁶³ Finally, three guidelines support the use of topical capsaicin, with one guideline stating that this topical agent is not effective or has discrepant results.^{158 160-163} One guideline supports topical lidocaine,¹⁶⁰ but this is not mentioned in many guidelines.

Global considerations

Although the Global Burden of Disease Study and the International Diabetes Federation systematically track prevalence of diabetes worldwide, the prevalence and severity of complications of diabetes, including DPN, are not as reliably assessed.²³² As noted, the lack of standard diagnostic criteria for DPN introduces substantial study heterogeneity, which complicates the calculation of pooled DPN incidence and prevalence rates across clinical sites and studies.

A few studies have attempted to assess DPN prevalence across countries. The INTERPRET-DD Study was a 14 country study examining the prevalence of diabetes and depression among 2733 participants obtaining outpatient care for type 2 diabetes.²³³ As part of this work, diabetes complications, including signs and symptoms of DPN, were abstracted from medical records at baseline and one year follow-up.²³⁴ On the basis of chart review, the study found an overall DPN prevalence of 26.7%, with significant variation between study sites (from 0.6% in Kenya to 79.6% in Ukraine).⁵⁸ Primary assessment of DPN symptoms and examination findings via standardized measures such as the MNSI overcome limitations related to chart abstraction. As shown in table 4, using these measures, prevalence of DPN has ranged from 8.5% (younger adults with type 1 diabetes in the US²³⁵) to 81.1% (adults with type 2 diabetes in Mexico; DOI link in table). Although five of the 22 studies included large, well phenotyped patient cohorts,^{43 44 46 61} most were single center studies with a small sample size.

GLOSSARY OF ABBREVIATIONS

- ADA—American Diabetes Association
- AGE—advanced glycation end products
- BMI—body mass index
- CBT—cognitive behavioral therapy
- DPN—diabetic peripheral neuropathy
- DRG—dorsal root ganglia
- GWAS—genome-wide association studies
- HbA_{1c}—hemoglobin A_{1c}
- HDL—high density lipoprotein
- IENFD—intraepidermal nerve fiber density
- LDL—low density lipoprotein
- MNSI—Michigan Neuropathy Screening Instrument
- MRI—magnetic resonance imaging
- mTCNS—modified Toronto Clinical Neuropathy Score
- NCS—nerve conduction studies
- NFL—neurofilament light chain
- SDOH—social determinants of health
- SNP—single nucleotide polymorphism
- SNRI—serotonin norepinephrine reuptake inhibitor

QUESTIONS FOR FUTURE RESEARCH

- What non-medical factors contribute to the development and severity of diabetic peripheral neuropathy (DPN)?
- Can multi-omics approaches identify new biomarkers with higher cellular specificity for DPN?
- Will studies that include DPN specific outcomes show that new drugs for diabetes (such as glucagon-like peptide-1 receptor agonists, sodium-glucose cotransporter 2 inhibitors, and dipeptidyl peptidase 4 inhibitors) help to slow the progression of DPN?
- Can new animal models that more closely recapitulate the contribution of metabolic syndrome to the development of DPN in type 2 diabetes allow for more efficient therapeutic development for DPN?

HOW PATIENTS WERE INVOLVED IN THE CREATION OF THIS ARTICLE

While writing this manuscript, we consulted a patient who is living with severe diabetic peripheral neuropathy (DPN) and helped to review the article and provide feedback. We asked the patient about what clinical and scientific aspects she thought were most pressing for patients affected by DPN. Her comments emphasized the need to educate patients about the high prevalence of DPN, the broad phenotypic spectrum of the disease, and the varied metabolic risk factors to ensure that patients know that the condition is common, differs in its presentation, and is not necessarily in their control. She believed that more education about the causes of DPN would alleviate the guilt that patients feel when DPN progresses despite their efforts to modify their lifestyles. She also stated that clinicians should continue to stress the potential for negative impact on mood and work productivity related to the side effects of neuropathic pain medication and to instruct patients to communicate such side effects as soon as they become aware of them. Finally, the patient expressed the need for broader patient education on treatment approaches for neuropathic pain that are not yet substantiated by scientific studies. She appreciated the scientific and medical community's efforts to develop more effective treatments for DPN and alleviate patients' suffering.

Therefore, the external validity of these findings may be limited. Only three studies conducted in France, Scotland, and rural Ecuador included population based recruitment.^{61 236 237} In France and Scotland, prevalence of DPN was 11% and 13%, respectively, whereas it was 59% in Ecuador. These differences may reflect underlying severity and duration of diabetes, as well as SDOH, such as access to care. Additional large, population based studies using standardized protocols are warranted to assess the impact of these risk factors on DPN prevalence.

Conclusions

The risk to global health posed by the rapidly increasing prevalence of DPN is enormous, and

continued research efforts are urgently needed to change the outlook for people living with this disabling condition. Encouragingly, the past decade has seen notable advances in defining the pathogenesis of DPN, as well as its biological and environmental risk factors. Specifically, the importance of axoglial interactions and metabolic derangements beyond hyperglycemia, such as dyslipidemia and insulin resistance, has opened new avenues for therapeutic investigation. At the same time, these insights have highlighted the clinical and biological complexity and heterogeneity of DPN, underscoring the need for more nuanced preclinical models and more precise approaches to diagnosis and treatment of DPN. Another theme that clearly resonates through every aspect of DPN research, from studies of prevalence and risk factors including SDOH to natural history studies and clinical trials, is the need to use more uniform definitions and measures of progression in DPN. Attaining this level of standardization will require multicenter consortiums and cross training of investigators in DPN outcomes. Finally, to take advantage of the rapid progress in the DPN field and expedite therapeutic development, new disease specific biomarkers will need to be developed and tested in rigorous longitudinal studies.

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